

Problem Statement

Reputation systems are widely used in online platforms to measure trustworthiness. A reputation system collects information about past behavior and converts it into a score that helps others decide whether to trust a user.

BUT, most reputation systems suffer from two major limitations:

1. Static Reputation→

Reputation scores often accumulate over time and do not decrease when a user becomes inactive. If a user stops participating for a long period, the system should reflect that inactivity. Without time based adjustments, the score may no longer represent current reliability.

2. Lack of Explainability→

In many systems, users can see that a reputation score has changed, but they cannot clearly understand why it changed. Blockchain smart contracts execute rules automatically, but they typically do not provide structured, human readable explanations for state updates. This reduces user confidence and system transparency.

Even though blockchain allows for immutability and transparency of transactions, many blockchain based reputation systems do not fully utilize these properties to create time aware and explainable trust mechanisms.

Therefore, there is a need for a simple and deterministic smart contract system that:

- Applies **time-based** reputation decay,
- Updates reputation through defined **interaction rules**,
- Provides **clear explanations** for every reputation change,